

COLLABORATIVE TEACHING PROJECT

Distributed Topics

Collaborative Lecture Project

CS 4504 Parallel and Distributed Computing | Instructor: Kun Suo

45 STUDENTS	12 TOPIC GROUPS	45-60 MINUTE LECTURE	100 PROJECT POINTS
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1. Project purpose

Each group will become the class expert on one distributed-computing topic. Your group will self-study the topic, redesign the supplied instructional material, build or demonstrate a technical example, and teach the topic through a recorded lecture.

Central expectation: Do not merely summarize the existing slides. Teach the topic in your own structure, verify the technical content, add meaningful new material, and demonstrate that every member understands the part they present.

Learning outcomes

- Explain a distributed-computing concept accurately and at an appropriate technical depth.
- Connect algorithms and architectures to realistic systems, tradeoffs, and failure modes.
- Create and test a reproducible code demonstration or technical walkthrough.
- Collaborate on a coherent lecture while showing clear individual ownership.
- Use sources and AI tools responsibly, transparently, and critically.

2. Teams and topic assignments

The class has 45 students and 12 topics. The class will use nine groups of four students and three groups of three students. The instructor will post the group roster and topic assignment in D2L.

Group	Assigned topic	Course theme
1	Distributed mutex	Coordination
2	Distributed election	Coordination
3	Distributed consensus	Coordination
4	Distributed locks	Coordination
5	Centralized structure	Management
6	Decentralized structure	Management
7	MapReduce	Computation
8	Spark	Computation
9	CAP theorem	Storage
10	Distributed storage	Storage
11	Publish and subscribe	Communication
12	Message queue	Communication

3. Fair division of sections and speaking time

Before building the lecture, divide the topic into N meaningful sections or subtopics. If the group has M students, distribute the sections as follows:

1. If N is divisible by M , each student presents N/M sections.
2. If N is not divisible by M , let $q = \text{floor}(N/M)$ and $r = N \bmod M$. Then r students present $q + 1$ sections and the remaining $M - r$ students present q sections.
3. Section count alone is not enough. Balance the assignment by estimated speaking time, technical difficulty, research effort, and demo responsibility.
4. A code demo may count as one section when it includes design, implementation, execution, and explanation.

Speaking-time target: For a four-person group, each student should present about 11-15 minutes. For a three-person group, each student should present about 15-20 minutes. Every student must speak for at least 10 minutes, and the complete lecture must remain between 45 and 60 minutes.

Required first slide: responsibility map

Slide 1 must identify the topic, group number, every member, assigned sections, and estimated speaking time. Highlight ownership with a distinct color for each member, and use the same small color marker or speaker name on that student's slides throughout the deck.

Member	Highlight	Responsible sections	Est. time
Student A	Blue	Motivation; definitions	12 min
Student B	Green	Architecture; core algorithm	14 min
Student C	Orange	Worked example; tradeoffs	12 min
Student D	Purple	Code demo; summary	12 min

4. Required lecture content

Organize the lecture as a coherent lesson rather than a collection of independent mini-presentations. The exact sections may vary by topic, but the final lecture must include:

- **Motivation and learning objectives.** Explain the problem and what the class should learn.
- **Foundations and terminology.** Define the necessary system model, assumptions, and vocabulary.
- **Core mechanism.** Teach the architecture, protocol, algorithm, or theorem step by step.
- **Worked example.** Trace a concrete example with messages, states, data flow, or calculations.
- **Real-system connection.** Discuss at least one real framework, platform, service, or published system.
- **Technical demonstration.** Run and explain code, a simulation, a benchmark, or a reproducible system walkthrough.
- **Tradeoffs and failure behavior.** Address performance, scalability, consistency, availability, fault tolerance, or security as applicable.
- **Summary and knowledge check.** End with key takeaways and at least three questions for classmates.

5. Existing slides and originality

The supplied slides are reference material, not a finished submission. You may use them to understand the expected scope and to identify important concepts, but your group must produce a meaningfully revised lecture.

- At least 50% of the substantive teaching content must be new or substantially redesigned by the group.
- No supplied slide may be copied unchanged. A substantially revised slide must improve the explanation, structure, evidence, visualization, example, or technical accuracy - not merely change colors or fonts.
- Do not reproduce a long sequence of the supplied deck. Reorganize the narrative around your own learning objectives and teaching plan.
- Cite the supplied slides when adapting them, and cite every external diagram, dataset, quotation, benchmark, or nontrivial claim.

6. Code demonstration requirements

The lecture must contain a technical demonstration lasting approximately 5-12 minutes. The demo should make an important idea visible rather than simply showing source code on screen.

- State the question the demo answers and connect it to the lecture concept.
- Show the architecture, inputs, commands, and expected behavior before execution.
- Run the demo and interpret the output. Do not treat successful execution as self-explanatory.
- Include at least one meaningful variation, comparison, failure, or edge case when appropriate.
- Provide source code plus a README with prerequisites, setup steps, commands, expected output, and known limitations.
- Prepare a short backup recording or screenshots in case a live run fails during recording.

Topic family	Possible demonstration direction
Coordination	Simulate mutual exclusion, leader election, consensus, or a distributed lock under delay/failure.
Management	Compare centralized and decentralized control, discovery, scheduling, or failure handling.
Computation	Run a MapReduce or Spark workload and explain partitioning, shuffle, parallelism, and timing.
Storage	Demonstrate replication, partition behavior, consistency tradeoffs, or a small distributed key-value store.
Communication	Build producer/consumer or publish/subscribe messaging and examine delivery, buffering, and failures.

7. Recorded lecture requirements

- Total duration: 45-60 minutes. Lectures outside this range may lose points unless approved in advance.
- Every member must narrate and present the sections listed on Slide 1. AI-generated voice or an AI avatar may not replace a student's presentation.
- Use a readable screen resolution of at least 720p, clear audio, and a stable view of slides and demonstrations.
- Add captions and review the automatic transcript for technical names, formulas, and code terms.
- Use clear handoffs between speakers. The final recording should feel like one lecture, not unrelated clips.
- Edit out extended setup delays, private information, broken attempts, and long silent periods.

8. Responsible use of AI

AI tools are allowed for code writing and file creation, including outlines, diagrams, images, slide drafts, captions, datasets, and formatting. AI use does not transfer responsibility for correctness to the tool.

- Verify every AI-generated technical claim against reliable sources.
- Compile, run, test, and understand all AI-assisted code before presenting it.
- Do not submit fabricated citations, fake benchmark results, invented system behavior, or untested output.
- Include an AI Use Appendix listing the tool, what it produced or changed, what the group retained, and how the result was verified.
- Each student must be able to answer questions about any AI-assisted material in that student's assigned section.

9. Deliverables

Item	Required contents
Editable slide deck	PPTX with the required responsibility map, speaker markers, citations, and AI appendix.
Slide export	PDF version of the final slides.
Recorded lecture	One 45-60 minute video file or instructor-approved streaming link.
Demo package	Source code, configuration/data files, and a reproducible README.
Contribution record	One page listing each member's research, slides, code, editing, and presentation work.
Peer evaluation	Each student submits the confidential individual evaluation in D2L.

10. Workflow and checkpoints

1. **Topic analysis:** Identify N sections, prerequisites, learning objectives, and likely demo.
2. **Ownership plan:** Submit the Slide 1 responsibility map and individual speaking-time estimates.
3. **Content review:** Share the revised outline, sources, and a list of material that is new or substantially changed.
4. **Demo checkpoint:** Run the code successfully from the README on a clean environment or VM.
5. **Rehearsal:** Record a short test, check timing, and obtain peer feedback on clarity and transitions.
6. **Final submission:** Submit all deliverables by the dates posted in D2L.

Submission naming

- G##_Topic_Slides.pptx
- G##_Topic_Slides.pdf
- G##_Topic_Lecture.mp4 (or link document)
- G##_Topic_Demo.zip (or instructor-approved repository link)
- G##_Topic_Contributions.pdf

One designated group member submits the group files. Every student submits the confidential peer evaluation separately.

11. Grading rubric (100 points)

Criterion	Pts	Evidence of strong performance
Technical accuracy and depth	25	Correct definitions, assumptions, mechanisms, examples, and analysis; appropriate depth for CS 4504.
Instructional design and organization	10	Clear objectives, coherent narrative, balanced sections, effective examples, and strong summary.
Originality, sources, and AI transparency	10	Meaningful revision beyond supplied slides; reliable citations; complete AI Use Appendix.
Code demonstration	15	Relevant, reproducible, tested, explained, and connected to the topic; useful variation or failure case.
Slides and visual communication	10	Readable, consistent, accessible, correctly labeled, and visually supportive of the explanation.
Recorded lecture and team integration	10	45-60 minutes, clear audio/video, reviewed captions, good transitions, professional pacing.
Individual mastery and delivery	15	Clear explanation, appropriate speaking time, command of assigned material, and ability to answer questions.
Contribution and professionalism	5	Reliable teamwork, accurate responsibility map, complete contribution record, and timely submission.
TOTAL	100	Group components: 80 points Individual components: 20 points

12. Final quality checklist

- Slide 1 clearly maps every section to a highlighted student owner.
- Speaking time is balanced and every student presents at least 10 minutes.
- The lecture is 45-60 minutes and includes a 5-12 minute technical demo.
- At least 50% of substantive content is new or substantially redesigned.
- No supplied slide is copied unchanged, and adapted material is cited.
- All code runs from the submitted README and the output is explained.
- Slides are readable, citations are complete, and captions are reviewed.
- The AI Use Appendix and contribution record are complete.
- All required files use the naming convention and open correctly.

Remember: The goal is not only to present information. Your group must teach the class well enough that classmates can understand, apply, and evaluate the topic after watching the lecture.